derivative rules

(sin csc −
cos − sec
tan 2 cot −2)

$\frac{d}{dx}\sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx}\cos^{-1}(x) = -\frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx}\frac{1}{1+x^2} = \frac{1}{1+x^2}$

5.5 HW

① $\int (6-5x)^9 dx$ $u = 6-5x$, $du = -5dx$

$= -\frac{1}{5}\int u^9 du = -\frac{1}{5}\frac{u^{10}}{10} = -\frac{1}{50}(6-5x)^{10}$

② $\int \frac{z^6}{z^7+7} dz$ $u = z^7+7$, $du = 7z^6 dz$, $\frac{1}{7}du = z^6 dz$

$\frac{1}{7}\int \frac{1}{u} du$

$\hookrightarrow = \frac{1}{7}\ln|x| \Rightarrow \frac{1}{7}\ln|z^7+7| + C$

③ $\int x\sqrt{8-x^2}\,dx$ $u = 8-x^2$, $du = -2x\,dx$, $-\frac{1}{2}du = x\,dx$

$-\frac{1}{2}\int u^{1/2} du$

$\hookrightarrow -\frac{1}{2}\left(\frac{2}{3}u^{\frac{3}{2}}\right) = -\frac{1}{3}(8-x^2)^{3/2}$

④ $\int \frac{x^6}{(x^7+9)^2} dx$ $u = x^7+9$, $du = 7x^6 dx$

$\frac{1}{7}\int u^{-2} du = \frac{1}{7}\frac{u^{-1}}{-1} = -\frac{1}{7u} \Rightarrow -\frac{1}{7(x^7+9)} + C$

⑤ $\int \frac{z^2}{\sqrt[3]{6+z^3}}$ $u = 6+z^3$, $du = 3z^2 dz$, $\frac{1}{3}du = z^2 dz$

$\frac{1}{3}\int \frac{1}{u^{1/3}} = \frac{1}{3}\int u^{-1/3} = \frac{1}{3}\cdot\frac{3}{2}u^{2/3} = \frac{1}{2}(6+z^3)^{2/}$

⑥ $\int e^{3x}\sin(e^{3x}) dx$ $u = e^{3x}$, $du = e^{3x}\cdot 3\,dx$, $\frac{1}{3}du = e^{3x}dx$

$\frac{1}{3}\int \sin u\,du$

$= \frac{1}{3}\left(-\cos(e^{3x})\right) + C$

⑦ $\int \frac{24x^2+48x}{x^3+3x^2+1} dx$ $u = x^3+3x^2+1$, $du = (3x^2+6x)dx$

$\int \frac{8(3x^2+6x)}{x^3+3x^2+1} dx = 8\int \frac{1}{u} du$

$= 8\ln|x^3+3x^2+1| + C$

⑧ $\int \frac{(\ln(5x))^{36}}{x} dx$ $u = \ln(5x)$, $du = 5\cdot\frac{1}{5x}dx$, $du = \frac{1}{x}dx$

$\int u^{36} = \frac{(\ln(5x))^{37}}{37}$

⑨ $\int (x^3+5)(x^4+20x+1)^4 dx$ $u = x^4+20x+1$, $du = 4x^3+20dx$, $du = 4(x^3+5)dx$, $\frac{1}{4}du = x^3+5\,dx$

$\frac{1}{4}\int u^4 du$

$= \frac{1}{4}\frac{u^5}{5} = \frac{1}{20}(x^4+20x+1)^5 +$

⑩ $\int e^{\tan 9x}\sec^2 9x\,dx$ $u = \tan 9x$, $du = \sec^2 9x\cdot 9$, $\frac{1}{9}du = \sec^2 9x\,dx$

$\frac{1}{9}\int e^u du$

$\frac{1}{9}e^u = \frac{1}{9}e^{\tan 9x} + C$

⑪ $\int \frac{\tan^{-1}(5x)}{1+25x^2}$ $u = \tan^{-1}(5x)$, $du = \frac{5}{1+25x^2}dx$

$\frac{1}{5}\int u\,du = \frac{1}{10}u^2 = \frac{1}{10}(\tan^{-1}5x)^2$

⑫ $\frac{\cos(\sqrt{5t})}{\sqrt{5t}} dt$ $u = \sqrt{5t} = \sqrt{5}\,t^{1/2}$, $du = \frac{\sqrt{5}}{2}\cdot\frac{1}{\sqrt{t}}dt$

$\frac{2}{5}\int \cos u\,du$ $\frac{2\sqrt{t}}{\sqrt{5}}du = dt$

$= \frac{2}{5}\sin(\sqrt{5t}) + C$

$du\ \frac{2\sqrt{t}}{\sqrt{5}}\times\frac{\sqrt{5}}{\sqrt{5}} = \frac{2\sqrt{5}\sqrt{t}}{5}du = dt$

$\frac{2}{5}du = \frac{dt}{\sqrt{5t}}$

⑬ $\int \sin(39t)\sec^2(\cos(39t))\,dt$

$u = \cos(39t)$, $du = -\sin(39t)\cdot 39$, $-\frac{1}{39}du = \sin(39t)dt$

$-\frac{1}{39}\int \sec^2(u)du$

$-\frac{1}{39}\tan((\cos)(39t)) + C$

sin cc −
cos − sec
tan 2 cot −2)

⑭ $\int_0^1 x^4(1+2x^5)^5 dx$ $u = 1+2x^5$, $du = 10x^4 dx$, $\frac{1}{10}du = x^4 dx$

$\frac{1}{10}\int_0^1 u^5 du$

$= \frac{1}{10}\left(\frac{u^6}{6}\right)\Big|_0^1 = \frac{1}{10}\left(\frac{(1+2x^5)^6}{6}\right)\Big|_0^1$

$= \frac{1}{10}\cdot\frac{728}{6} = \frac{728}{60}$

⑮ $\int_e^{e^9} \frac{1}{x\sqrt{\ln x}} dx$ $u = \ln x$, $du = \frac{1}{x}dx$

$\int_e^{e^9} u^{-1/2} du = \int_e^{e^9} 2\sqrt{u} = \int_e^{e^9} 2\sqrt{\ln x}\,dx$

$= 2\sqrt{\ln(e^9)} - 2\sqrt{\ln}$

⑯ $\int_1^6 \frac{e^{1/x}}{x^2} dx$ $u = x^{-1}$, $du = -\frac{1}{x^2}dx$, $-du = \frac{1}{x^2}dx$

$-\int_1^6 e^u du = -e^u\Big|_1^6 = -(e^{1/x})\Big|_1^1 = -(e^{1/6} - e^1)$

$= e - e^{1/6}$

⑰ $\int \frac{3+3x}{1+x^2} dx$

$3\int \frac{1}{1+x^2} + 3\int \frac{x}{1+x^2}$ $u = 1+x^2$, $du = 2x\,dx$, $\frac{1}{2}du = x\,dx$

$3\tan^{-1}(x) + \frac{3}{2}\int \frac{1}{u}du$

$3\tan^{-1}(x) + \frac{3}{2}\ln|1+x^2|$

⑱ $\int x(7-x)^7 dx$ $u = 7-x$, $du = -1dx$, $-du = dx$ $x = 7-u$

$-\int (7-u)u^7 du$

$\int (u-7)u^7 du \Rightarrow \int u^8 - 7u^7 \Rightarrow \frac{u^9}{9} - \underline{7u^8}$

⑲ $\int \frac{x}{\sqrt[4]{x+3}} dx$ $u = x+3$, $du = dx$ $x = u-3$

$\int \frac{u-3}{\sqrt[4]{u}} = \int \frac{u}{u^{1/4}} - \int \frac{3}{u^{1/4}} \Rightarrow \int u^1\cdot u^{-1/4} - 3\int u^{-1/4}$

$\int u^{3/4} - 3\int u^{-1/4}$

$\frac{4u^{7/4}}{7} - \cancel{3}\cdot\frac{4}{\cancel{3}}u^{3/4}$

$\frac{4}{7}u^{7/4} - 4u^{3/4} + C$

⑳ $\int x^3\sqrt{x^2+26}\,dx$ $u = x^2+26$, $du = 2x\,dx$, $\frac{1}{2}du = x\,dx$ $x^2 = u-26$

$\int x^2\cdot x\sqrt{x^2+26}\,dx$

$\frac{1}{2}\int (u-26)u^{1/2} du$

$\frac{1}{2}\int u^{3/2} - 26u^{1/2} = \frac{1}{2}\cdot\frac{2}{5}u^{5/2} - \frac{26}{2}\cdot\frac{2}{3}u^{3/2}$

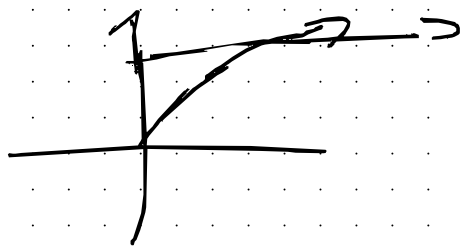
$\frac{1}{5}u^{5/2} - \frac{52}{6}u^{3/2} + C$

6.2 HW

1) $V = \int_4^{16} \pi \left(\frac{5}{x}\right)^2 dx$

2)

$y = \sqrt{2x}$



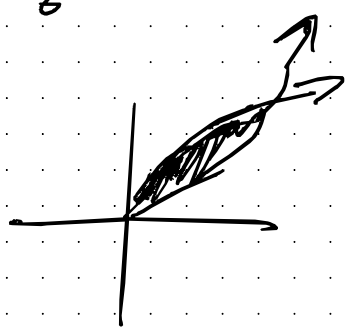
$x = \frac{y^2}{2}$

$V = \int_0^5 \pi \left(\frac{y^2}{2}\right)^2 dy$

3)

$y = x^5$

$y = \sqrt{x}$

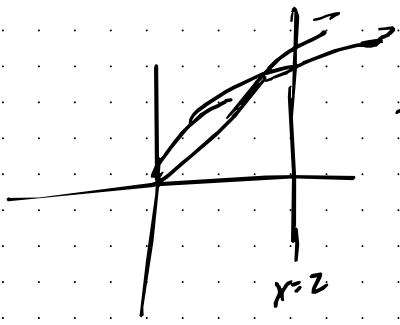


$V = \int_0^1 \pi \left([x^5]^2 - [\sqrt{x}]^2 \right) dx$

not $(r_1 - r_2)^2$

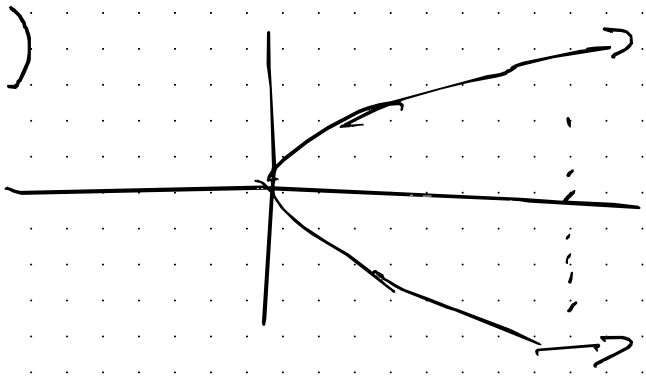
4) $y = x$

$y = \sqrt{x}$



$2\pi(2-x)(\sqrt{x}-x)$

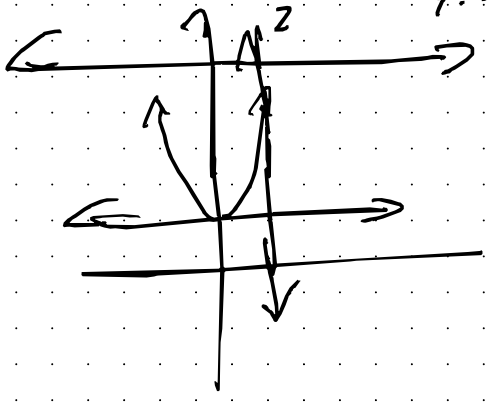
5)



$V = \pi \int_{-1}^{1} (9 - 9y^2)^2$

6)

$y = \frac{x^3}{2} + 3$ $\quad y = 7 \quad x = 4$



$V = \pi \int_0^4$

7)